# Dr. Soly Mathew - Curriculum Vitae

Materials Scientist | Flexible Sensor Researcher | Computational Materials & AI
ORCID: 0009-0001-2480-4682 | Email: soly.mathew@vit.ac.in | Web: soly-mathew.github.io

## ACADEMIC & RESEARCH APPOINTMENTS

Scientific Officer - Indian Institute of Heritage, Delhi (Jul 2026 - Present)
Dept of Art Conservation: Scientific examination, characterization & degradation analysis

Project Associate - Indian Institute of Technology Jodhpur (IIT Jodhpur) (Oct 2025 - Jun 2026)
R&D on functional nanocomposites, sensor design & electromechanical characterisation

Assistant Professor (Adhoc) - NSS College of Engineering, Palakkad (Aug 2025 - Oct 2025)
Undergraduate Engineering Physics curriculum delivery, laboratory instruction & mentoring

## EDUCATION

PhD in Materials Science - Vellore Institute of Technology (VIT), India (2020 - Present)
Focus: Flexible capacitive pressure sensors, polymer nanocomposites, ternary dielectrics

## PEER-REVIEWED JOURNAL PUBLICATIONS

1. Smart Materials and Structures (2026) - DOI: 10.1088/1361-665X/ae8228
2. Journal of Applied Polymer Science (2026) - DOI: 10.1002/app.70439
3. Flexible and Printed Electronics (2025) - DOI: 10.1088/2058-8585/adc81e
4. Sensors International (2025) - DOI: 10.1016/j.sintl.2025.100330
5. IEEE Access (2025) - DOI: 10.1109/ACCESS.2025.3577066

## CORE SCIENTIFIC & COMPUTATIONAL COMPETENCIES

Materials: Polymer Nanocomposites, PDMS, PVDF, ZnO, TiO2, ZrO2, Graphene Oxide, Carbon Black
Fabrication: Spin Coating, Solution Casting, Cleanroom Thermal Curing, Vacuum Degassing
Characterisation: FESEM, XRD, FTIR, BDS Broadband Dielectric Spectroscopy, LCR Testing
Computational & AI: Python 3, NumPy, SciPy, scikit-learn, PyTorch, LangGraph, Git, pytest

## TARGET INSTITUTIONAL COLLABORATION

Danish & European Research Institutes: DTI, FORCE Technology, Coloplast, DTU, SDU, AU